In the Design of Things

Universities all over the US have started greener initiatives. Carnegie Mellon University has started a campus-wide programme called Green Design Initiative. The programme works towards environmentally conscious engineering and production. The University has partnered with industrial corporations, foundations and government agencies to work on advanced research methodologies for a sustainable development.

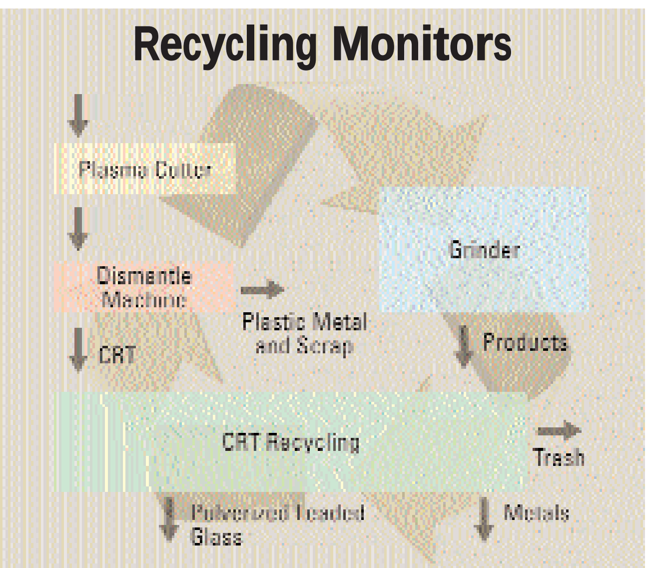
Green Design Initiative is doing research in specialised areas to aid the electronics industry to incorporate environmentally sound technologies in the production of their equipments. Green Design also encourages students who are interested in environment-oriented courses.

The consortium led by the Carnegie Mellon University does research on recycling computers and has identified that a large amount of electronic debris comes from the usage of computers by businesses. Studies have been conducted by the university for making a statistical analysis of how many computers are being dumped in landfills and how many have been recycled in the recent years. The research programme aims to reduce environmental damage by lowering hazardous material discharge, minimising the use of non-renewable resources and by reducing the usage of renewable resources to sustainable limits.

chemicals might leach into the ground water. The best solution therefore is to recycle them. Monitors, keyboards, motherboards and other peripherals such as floppies and CDs are all recyclable.

Let's take the case of monitors. Metals present in the printed circuit board and inside the housing of the monitor are recycled. The housing itself is shredded and used to make a special material, which is used to fill potholes. Most monitors have a cover of lead over the screen, which can be reused. Flat panel monitors have mercury lamps, which are used to backlight the screen. One of the most hazardous metals, it can be reused as a raw material for manufacturing newer flat panels.

Recycling monitors is what Total Reclaim, a Seattle-based company does. The plastic casing that covers a monitor is sent to a processing zone to be pelletised. These pellets are then used as feedstock for newer components or casing production. The CRTs are smelted and reused for making newer monitors. R3PC is another organisation that has entered into monitor recycling. Here, the monitor's casing is sliced out using a plasma cutter and is recycled. CRTs are pulverised and separated into leaded glass, metals and trash. All other parts are ground and sold to recyclers, who use these as raw materials to make new products.



The monitor casing is sliced using a plasma cutter, and is then recycled in different processing zones. The CRTs are pulverised and separated into glass, metals and trash before recycling

Infographic: Shyam Shirsekar

Separating the chaff

But what about a motherboard, you ask? A typical motherboard will have copper, silver and gold besides stains of heavy metals such as antimony, chromium, zinc, tin and lead.

California-based Fox Electronics recycles the most advanced processors to the cheapest component. The resistors, capacitors and other components are removed by de-soldering them from the motherboard. The chemically treated board is ripped off its ICs and is smelted again for producing powder, which can be reused to make motherboards and other PCBs.

Keyboards are cut along the seam using robots and are then separated into plastic and metal board. The plastic is melted and is turned into pellets for further use.

Going to CD

Computer peripherals such as floppies, CDs and others can be recycled similarly. According to

Safe Disposals

Although the CD has been a major invention as a portable storage media, disposal of corrupt and unsold CDs is an issue. Most CDs are manufactured from polymers, which are difficult to dispose.

A UK-based company, Polymer-Reprocessors Limited (PRL), is working towards recycling CDs and their jewel cases for making newer materials. PRL has created a system to recycle CDs and their cases for use in newer applications. Some of Europe's largest international music companies have joined hands with PRL in this environment friendly approach towards recycling CDs and their cases.

The CDs, along with their packing are separated into components such as paper, plastic, CDs and cases. The cases are granulated and fed into an extrusion system fitted with a laser for the removal of any contaminants. This process produces high quality crystal polystyrene pellets, which are used in making artificial wood, insulating foams or even CD cases once again. The discs are recycled using a specially patented recycling machine, which removes the paint, aluminium and eventually the data. The cleaned disc is granulated and the outcome is high-quality injection moulding grade polycarbonate, which is used in a multitude of applications—in the production of burglar alarms, streetlights, lenses, etc.

PRL is also working towards recycling DVDs, video cassettes, audio cassettes and tapes and computer reels.